Brief Overview
Artificial General Intelligence (AGI) could transform science and help solve some of humanity's biggest challenges but measuring progress toward it remains difficult.

Today's AI models can achieve strong benchmark scores, often by recognizing patterns or recalling training data rather than demonstrating true reasoning. This makes it hard to tell when a model is actually solving a new problem.

First Results/Outputs:

```
Tasks redefined successfully - no more assert_true error.
Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI


=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy:      0.891
Mean Confidence:       100.0%
Brier Score:           0.1087  (lower = better calibrated)
ECE:                   0.0000  (lower = better calibrated)

Per-Tier Breakdown:
      accuracy  mean_confidence  mean_brier   n
tier
1        1.000            100.0       0.000  10
2        1.000            100.0       0.000  12
3        0.833            100.0       0.167  12
4        0.750            100.0       0.250  12

Overconfidence Gap:    0.109
Composite Calibration Score: 0.8913

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy:        0.870
Forecast Accuracy:     0.870  (correctly predicted own outcome)
Error Recall:          0.000  (fraction of own errors predicted in advance)
False Positive Rate:   0.130  (thought correct, was wrong)

Per-Tier:
      forecast_accuracy  accuracy  error_recall
tier
1                 0.800     0.800           0.0
2                 1.000     1.000           0.0
3                 0.917     0.917           0.0
4                 0.750     0.750           0.0
```

Naive Baseline (always predict CORRECT): 0.870
Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate: 1.000 (caught planted wrong answers)
False Flag Rate: 0.000 (wrongly flagged correct answers)
Overall Confabulation: 0.000 (agreed with wrong answers)

Error Detection by Tier:

tier	error_detection_rate	n
2	1.0	12
3	1.0	12
4	1.0	12

Anti-Confabulation Score: 1.0000

==
COMPOSITE METACOGNITION SCORE
==
Calibration Component (40%):   0.8913
Forecasting Component (30%):   0.5000
Confabulation Component (30%): 1.0000

COMPOSITE SCORE: 0.8065
==

Interpretation:
> 0.80: Excellent metacognitive ability
0.60-0.80: Good but with identifiable gaps
0.40-0.60: Moderate — significant calibration or forecasting issues
< 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8065

Second Results/Outputs:

Tasks redefined successfully - no more assert_true error.
Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI

=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy: 0.913
Mean Confidence: 100.0%
Brier Score: 0.0870 (lower = better calibrated)

ECE: 0.0000 (lower = better calibrated)

Per-Tier Breakdown:

tier	accuracy	mean_confidence	mean_brier	n
1	1.000	100.0	0.000	10
2	1.000	100.0	0.000	12
3	0.917	100.0	0.083	12
4	0.750	100.0	0.250	12

Overconfidence Gap: 0.087
Composite Calibration Score: 0.9130

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy: 0.935
Forecast Accuracy: 0.935 (correctly predicted own outcome)
Error Recall: 0.000 (fraction of own errors predicted in advance)
False Positive Rate: 0.065 (thought correct, was wrong)

Per-Tier:

tier	forecast_accuracy	accuracy	error_recall
1	1.000	1.000	0.0
2	1.000	1.000	0.0
3	0.917	0.917	0.0
4	0.833	0.833	0.0

Naive Baseline (always predict CORRECT): 0.935
Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate: 1.000 (caught planted wrong answers)
False Flag Rate: 0.000 (wrongly flagged correct answers)
Overall Confabulation: 0.000 (agreed with wrong answers)

Error Detection by Tier:

tier	error_detection_rate	n
2	1.0	12
3	1.0	12
4	1.0	12

Anti-Confabulation Score: 1.0000

==
COMPOSITE METACOGNITION SCORE

```
================================================================
  Calibration Component (40%):   0.9130
  Forecasting Component (30%):   0.5000
  Confabulation Component (30%): 1.0000
  ──────────────────────────────────────────────────
  COMPOSITE SCORE:             0.8152
================================================================
```

```
Interpretation:
  > 0.80: Excellent metacognitive ability
  0.60-0.80: Good but with identifiable gaps
  0.40-0.60: Moderate — significant calibration or forecasting issues
  < 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8152
```

Third Results/Outputs

```
Tasks redefined successfully - no more assert_true error.
Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI


=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy:      0.913
Mean Confidence:       100.0%
Brier Score:           0.0870  (lower = better calibrated)
ECE:                   0.0000  (lower = better calibrated)

Per-Tier Breakdown:
```

tier	accuracy	mean_confidence	mean_brier	n
1	0.900	100.0	0.100	10
2	1.000	100.0	0.000	12
3	0.917	100.0	0.083	12
4	0.833	100.0	0.167	12

```
Overconfidence Gap:    0.087
Composite Calibration Score: 0.9130

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy:        0.913
Forecast Accuracy:     0.913  (correctly predicted own outcome)
Error Recall:          0.000  (fraction of own errors predicted in advance)
False Positive Rate:   0.087  (thought correct, was wrong)
```

Per-Tier:

tier	forecast_accuracy	accuracy	error_recall
1	1.000	1.000	0.0
2	1.000	1.000	0.0
3	0.917	0.917	0.0
4	0.750	0.750	0.0

Naive Baseline (always predict CORRECT): 0.913
Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate: 1.000 (caught planted wrong answers)
False Flag Rate: 0.000 (wrongly flagged correct answers)
Overall Confabulation: 0.000 (agreed with wrong answers)

Error Detection by Tier:

tier	error_detection_rate	n
2	1.0	12
3	1.0	12
4	1.0	12

Anti-Confabulation Score: 1.0000

==
COMPOSITE METACOGNITION SCORE
==
Calibration Component (40%): 0.9130
Forecasting Component (30%): 0.5000
Confabulation Component (30%): 1.0000

COMPOSITE SCORE: 0.8152
==

Interpretation:
> 0.80: Excellent metacognitive ability
0.60-0.80: Good but with identifiable gaps
0.40-0.60: Moderate - significant calibration or forecasting issues
< 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8152
NEW:
First Results/Outputs

Tasks redefined successfully - no more assert_true error.

Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI

=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy: 0.913
Mean Confidence: 100.0%
Brier Score: 0.0870 (lower = better calibrated)
ECE: 0.0000 (lower = better calibrated)

Per-Tier Breakdown:

	accuracy	mean_confidence	mean_brier	n
tier				
1	0.900	100.0	0.100	10
2	1.000	100.0	0.000	12
3	0.917	100.0	0.083	12
4	0.833	100.0	0.167	12

Overconfidence Gap: 0.087
Composite Calibration Score: 0.9130

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy: 0.891
Forecast Accuracy: 0.891 (correctly predicted own outcome)
Error Recall: 0.000 (fraction of own errors predicted in advance)
False Positive Rate: 0.109 (thought correct, was wrong)

Per-Tier:

	forecast_accuracy	accuracy	error_recall
tier			
1	0.900	0.900	0.0
2	1.000	1.000	0.0
3	0.917	0.917	0.0
4	0.750	0.750	0.0

Naive Baseline (always predict CORRECT): 0.891
Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate: 0.972 (caught planted wrong answers)
False Flag Rate: 0.000 (wrongly flagged correct answers)
Overall Confabulation: 0.022 (agreed with wrong answers)

Error Detection by Tier:

	error_detection_rate	n
tier		

```
2              0.917  12
3              1.000  12
4              1.000  12

Anti-Confabulation Score: 0.9722

============================================================
COMPOSITE METACOGNITION SCORE
============================================================
  Calibration Component (40%):   0.9130
  Forecasting Component (30%):   0.5000
  Confabulation Component (30%): 0.9722
  --------------------------------------------------
  COMPOSITE SCORE:               0.8069
============================================================

Interpretation:
  > 0.80: Excellent metacognitive ability
  0.60-0.80: Good but with identifiable gaps
  0.40-0.60: Moderate - significant calibration or forecasting issues
  < 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8069
```

Second Results/Outputs:

```
Tasks redefined successfully - no more assert_true error.
Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI


=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy:      0.957
Mean Confidence:       100.0%
Brier Score:           0.0435  (lower = better calibrated)
ECE:                   0.0000  (lower = better calibrated)

Per-Tier Breakdown:
      accuracy  mean_confidence  mean_brier   n
tier
1        1.000           100.0       0.000  10
2        1.000           100.0       0.000  12
3        0.917           100.0       0.083  12
4        0.917           100.0       0.083  12
```

```
Overconfidence Gap:    0.043
Composite Calibration Score: 0.9565

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy:        0.913
Forecast Accuracy:     0.913  (correctly predicted own outcome)
Error Recall:          0.000  (fraction of own errors predicted in advance)
False Positive Rate:   0.087  (thought correct, was wrong)

Per-Tier:
      forecast_accuracy  accuracy  error_recall
tier
1                 0.800     0.800           0.0
2                 1.000     1.000           0.0
3                 0.917     0.917           0.0
4                 0.917     0.917           0.0

Naive Baseline (always predict CORRECT): 0.913
Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate:     0.972  (caught planted wrong answers)
False Flag Rate:          0.000  (wrongly flagged correct answers)
Overall Confabulation:    0.022  (agreed with wrong answers)

Error Detection by Tier:
      error_detection_rate   n
tier
2                    0.917  12
3                    1.000  12
4                    1.000  12

Anti-Confabulation Score: 0.9722

============================================================
COMPOSITE METACOGNITION SCORE
============================================================
  Calibration Component (40%):   0.9565
  Forecasting Component (30%):   0.5000
  Confabulation Component (30%): 0.9722
  ----------------------------------------------------
  COMPOSITE SCORE:               0.8243
============================================================

Interpretation:
  > 0.80: Excellent metacognitive ability
```

0.60-0.80: Good but with identifiable gaps
0.40-0.60: Moderate - significant calibration or forecasting issues
< 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8243

Third Results/Outputs:

Tasks redefined successfully - no more assert_true error.
Running Calibration Blindspot Benchmark (CBB)
Track: Metacognition | Hackathon: Measuring Progress Toward AGI

=== TASK 1: CONFIDENCE CALIBRATION ===
Overall Accuracy: 0.935
Mean Confidence: 100.0%
Brier Score: 0.0652 (lower = better calibrated)
ECE: 0.0000 (lower = better calibrated)

Per-Tier Breakdown:

tier	accuracy	mean_confidence	mean_brier	n
1	0.900	100.0	0.100	10
2	1.000	100.0	0.000	12
3	0.917	100.0	0.083	12
4	0.917	100.0	0.083	12

Overconfidence Gap: 0.065
Composite Calibration Score: 0.9348

=== TASK 2: PRE-ANSWER ERROR FORECASTING ===
Model Accuracy: 0.891
Forecast Accuracy: 0.891 (correctly predicted own outcome)
Error Recall: 0.000 (fraction of own errors predicted in advance)
False Positive Rate: 0.109 (thought correct, was wrong)

Per-Tier:

tier	forecast_accuracy	accuracy	error_recall
1	0.800	0.800	0.0
2	1.000	1.000	0.0
3	0.917	0.917	0.0
4	0.833	0.833	0.0

Naive Baseline (always predict CORRECT): 0.891

Metacognitive Lift: +0.000

=== TASK 3: CONFABULATION DETECTION ===
Error Detection Rate: 0.972 (caught planted wrong answers)
False Flag Rate: 0.000 (wrongly flagged correct answers)
Overall Confabulation: 0.022 (agreed with wrong answers)

Error Detection by Tier:

	error_detection_rate	n
tier		
2	0.917	12
3	1.000	12
4	1.000	12

Anti-Confabulation Score: 0.9722

==
COMPOSITE METACOGNITION SCORE
==
Calibration Component (40%):   0.9348
Forecasting Component (30%):   0.5000
Confabulation Component (30%): 0.9722

COMPOSITE SCORE: 0.8156
==

Interpretation:
> 0.80: Excellent metacognitive ability
0.60-0.80: Good but with identifiable gaps
0.40-0.60: Moderate - significant calibration or forecasting issues
< 0.40: Poor - high confabulation or severe overconfidence

Final Composite Score: 0.8156

Comparison Cell Results/Outputs

Comparison dataset: 15 items across 4 tiers
MULTI-MODEL CBB COMPARISON
Running with kbench.llm and kbench.judge_llm
==

==
Testing: default_model
==
Accuracy: 0.867

Mean Confidence: 100.0%
Brier Score: 0.1333
Overconfidence Gap: 0.133
Error Recall: 0.000
Metacog Lift: +0.000
Partial CBB Score: 0.7090 (Tasks 1+2 only)

Calibration by Tier:

tier	is_correct	confidence
1	1.0	100.0
2	1.0	100.0
3	1.0	100.0
4	0.5	100.0

==
Testing: judge_model
==
Accuracy: 0.800
Mean Confidence: 100.0%
Brier Score: 0.2000
Overconfidence Gap: 0.200
Error Recall: 0.000
Metacog Lift: +0.000
Partial CBB Score: 0.6710 (Tasks 1+2 only)

Calibration by Tier:

tier	is_correct	confidence
1	1.00	100.0
2	1.00	100.0
3	1.00	100.0
4	0.25	100.0

==
MULTI-MODEL COMPARISON SUMMARY
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model	accuracy	mean_confidence	brier_score	error_recall	partial_cbb_score
default_model	0.867	100.0	0.1333	0.0	0.709
judge_model	0.800	100.0	0.2000	0.0	0.671

Key insight: Models with non-zero Error Recall demonstrate
genuine metacognitive self-knowledge above the naive baseline.

Note: Your main benchmark (46 items, 3 tasks) remains the
primary submission. This comparison demonstrates discriminatory power.